



Report v6 — SR 11-7 model validation: where the losses are, and what the record can prove

SR 11-7 asks whether each model does what it claims, on independent evidence. This review covers every model that publishes, ranks or trades an idea, on the clean-era record (ideas published from 2026-08-26). Prepared September 30, 2026, after the close; post-report updates from the same evening are in section 10.

Read this first. Section 10 supersedes parts of the body. **F-0 (journal regression) is fixed:** 18:28 ET, 470 rows load. The memory restarts in F-16 have not recurred since the 18:11 ET deploy. Loss Rules v1.1 and the ODTE sniper (watch mode) went live this evening and are **measuring**. **F-11 (the evidence score does not rank outcomes) is raised to High.**

Date: 2026-09-30, after the close. **Previous reports:**

- v1 and v2 (2026-09-24): replay ground truth; stop width identified as the #1 loss driver.
- v3: the $1.25 \times \text{ATR}$ stop floor.
- v4: UI and UX.
- v5: surface inventory.

Scope: every model that publishes, ranks or trades an idea on the platform, measured on the clean-era record (ideas published from 2026-08-26). All evidence was produced read-only against production. Nothing was written, deployed or restarted.

Evidence produced today:

- `research/loss-attribution.ts` -> `research/loss-attribution-results.json`
- `research/shakeout-review.ts` -> `research/shakeout-review-results.json`
- The companion write-up, `docs/LOSS_ATTRIBUTION_2026-09-30.md`, has every table.
- Replay results from `research/flush-reclaim-replay.ts` were supplied by the operator; they were not re-run for this report.

Status labels:

- **Validated:** independent evidence supports the claim.
- **Not supported:** the evidence contradicts the claim.
- **Measuring:** too few observations, or no independent test yet.

1. Summary

1. NEXUS ideas (unit-sized book)

- Record: 264 closed trades, 39.0% positive, net **-\$4,070** (-\$15.42 per trade).
- H1 (Aug 26 – Sep 12): -\$5,500 on n=218.
- H2 (Sep 13 – Sep 30): +\$1,429 on n=46.
- H2 is too small to call a turn.

2. The losses are concentrated, not spread evenly.

- Options published **after the close**: -\$4,516 (n=72).
- market_scanner 8–21 DTE call swings: -\$3,738 (n=86). The Aug 31 batch alone lost -\$7,634.
- Options published during regular hours made +\$1,758 (n=73).

3. Stops are not the main leak.

- 37% of stop-outs later turned green (17 of 46). Those shakeouts cost only -\$1,447.
- Holding every priceable stop-out to the end of its plan would have lost **more** (-\$5,081 vs -\$4,411).
- Contract-% stop rules were worse in both halves.
- The $1.25 \times \text{ATR}$ floor looks best, but only because of one trade that was published twice. Its status goes back to **measuring**.

4. The outcome record has known defects that distort the metrics. All are listed in section 7:

- 83% of exit timestamps are shared across symbols (batch sweeps);
- option stop exits are priced at the sweep, not at the stop;
- a date-parse bug expires every expiry-day option idea the evening before;
- the trigger observer triggers ideas whose stop is already breached;
- 374 of 844 ideas cannot be scored at all.

5. Quantinum Bot. Net **+\$103** on 40 closed fills. Run 2 made +\$4,908; **Run 3 (from Sep 24) is 0 for 9, -\$4,632.** Only quant-sourced and band-A signals are positive.

6. Operations.

- Production restarted **40 times today** on the memory limit: 20 web, 20 worker, between 11:40 ET and 6 PM ET.
- A deploy at 17:10 CT (c1980ccb) broke the NEXUS journal book: the desk loader throws `invalid regular expression`. **Critical; fix before the next session.**

2. Model inventory

#	Model	Purpose	Code	Output since 08-26	Status
M1	NEXUS engines: the idea producers	Publish trade ideas (entry, stop, target, contract)	see rows below	940 published ideas; 470 scorable	mixed
M1a	market_scanner	multi-setup equity/option scanner	server/market-scanner.ts -> trade-idea-ingestion.ts	426 ideas, 162 closed, -\$3,941	Not supported (H1 -\$5,018 on n=158; H2 +\$1,077 on only n=4)
M1b	quant	quant-engine ideas	server/quantitative-analysis-engine.ts	168 ideas, 55 closed, +\$293	Measuring (+\$5.33 per trade, 47% win rate)
M1c	gex_scanner	gamma-level ideas	server/gex-idea-scanner.ts (writes via storage.createTradeIdea, bypassing the ingestion gates)	140 ideas, 28 closed, -\$1,827	Not supported (negative in both halves)
M1d	flow	options-flow ideas	flow tape pipeline	31 ideas, 8 closed, +\$675	Measuring (n=8)
M1e	spx_session	SPX session scalps	server/spx-session-scanner.ts	153 ideas; 0 scorable (no entry premium)	Suspended since 09-24 (-0.27R in v2)
M1f	gex_magnet, swing_catcher, orb_scanner, crypto_engine	smaller producers	various	5 / 2 / 1 / 6 ideas	Measuring
M2	Conviction engine (gen_conviction_band S/A/B/C)	Ranks ideas at publish	server/convictions-engine.ts	bands on 235 closed desk trades	Not supported as a ranker: S -\$388.81 per trade (n=6), A +\$13.58, B +\$17.53, C -\$51.26
M3	Quantinum Bot (paper execution)	Takes signals, sizes them, manages exits (premium stops, trailing, gap_magnet)	server/quant-bot.ts, server/bot-ledger.ts	46 fills, 40 closed, +\$103	Run 2 +; Run 3 0/9 -> Not supported for non-quant signals
M4	ODTE desk (and the ODTE sniper, deployed 17:10 CT today)	Same-day index/ETF option ideas	server/zero-dte-desk*.ts, server/zero-dte-sniper*.ts	3 closed desk ideas, -\$72	Measuring. Invalid until the expiry-date parse is fixed (F-3)
M5	Sector ignition	Sector-rotation ignition ideas	server/sector-ignition.ts	2 ideas (today), 0 closed	Measuring
M6	Pre-market ideas (premarket_gap)	Pre-market gap ideas	server/premarket-ideas.ts	0 published in the window	Measuring (no output)
M7	GEX engine (levels, rankings, magnets)	Gamma exposure used by M1c, M1f and the UI	server/gex-*.ts	feeds M1c/M1f	Measuring. Its idea outputs (M1c) are not supported; no independent level-accuracy test this cycle
M8	Outcome tracker and trigger observer	Grade ideas (target/stop/expiry, exit time, exit premium)	server/performance-validator.ts, server/performance-validation-service.ts, shared/exit-hit-time.ts, oracle lifecycle	every outcome above	Defects found (F-1, F-2, F-3, F-5)

#	Model	Purpose	Code	Output since 08-26	Status
M9	Loss Rules v1 (target cap, time stop)	Cap targets at expected move; exit at half-horizon unless $\geq 0.5R$	server/loss-rules.ts, shared/loss-rules.ts	2 time stops in the window	Measuring. Mis-applied when holding_period is wrong (F-7)

3. Data lineage and quality

Input	Source	Status	Effect
Option chains and quotes	Tradier (primary)	Tradier key dead since 2026-09-30. CBOE and Alpaca are the fallbacks (server/alpaca-options.ts, the shared CBOE loader)	Entry/exit premiums and marks come from delayed or indicative fallbacks. Source attribution is per mark (journal-marks) but not per stored premium
Flow direction	options_flow_history	sentiment is 'unknown' on 100% of rows since 2026-08-27 (e.g. 253 of 253 on 09-30, 476 of 476 on 09-24)	Chain-snapshot flow carries no direction. Any flow "bias" or "tape flipped" logic built on it is unsupported
Underlying bars (for this report)	Yahoo chart API: 5-minute RTH bars since 08-19, daily bars for ATR	Complete for 262 of 264 trades (TRUMP and SUI missing)	Independent of the tracker
Option bars (for this report)	Alpaca v1beta1/options/bars, 1-hour	158 of 161 option trades have prints (after capping end 20 minutes before now; the plan rejects the latest 15 minutes)	Hourly closes are coarse; stated
Stored option premiums	trade_ideas.entry_premium / exit_premium	108 option ideas have no entry premium (all spx_session and 43 gex_scanner). 108 are resolved with no exit premium	374 of 844 ideas cannot be scored (44%)
Exit timestamps	trade_ideas.exit_date (America/Chicago)	433 of 522 share a stamp with another symbol. Only 33 carry an [exit-time] tag	Hour-of-exit and holding-time analyses before 09-30 are unreliable

4. Outcomes vs the honest baseline

- **Baseline (2026-08-26, post-fix):**
 - 42% win rate, +0.137R per trade, on n=88 decided trades (Wilson 95% CI 32–52%).
 - Statistically indistinguishable from zero.
- **Today, same clean era, larger n:**
 - **Barrier outcomes** across all engines: 29 targets vs 138 stops. That is **17.4% of barrier-decided ideas**. The rest (about 560) expired, were collapsed, or are open.
 - **Journal P&L**, unit-sized: 264 closed, 39.0% positive, -\$15.42 per trade. Average win +\$238.58 vs average loss -\$189.70, a payoff ratio of 1.26. At a 39% win rate that ratio is below break-even (1.56 would be needed).
 - Expectancy in R is **not** restated here. Option P&L is in contract dollars while stops are underlying levels, so a single R is not defined across the book.
- **Conclusion:** there is still no evidence of edge at the book level. The positive pockets are small:
 - RTH-published options: +\$1,758 (n=73);
 - quant: +\$293 (n=55);

- ODTE: +\$696 (n=15), but that result is invalid under F-3;
 - lotto: +\$1,042 (n=20).
- All of these are "measuring".

5. Today's findings

5.1 Loss attribution

Detail is in docs/LOSS_ATTRIBUTION_2026-09-30.md.

- After-close option publishing: -\$4,516 (n=72).
- market_scanner call swings: -\$3,738 (n=86).
- gex_scanner: -\$1,827 (n=28), negative in both halves.
- quant stock swings: -\$1,013 (n=8).
- The 15 worst trades are 57% of gross losses.
- The conviction band and the probability band do not rank outcomes.

5.2 Shakeouts

- **Stop-outs:** of 46 losing stop-outs, 6 later hit the original target and 11 turned green without the target (37%).
 - By engine: quant 67%, flow 67%, gex_scanner 25%, market_scanner 25%.
 - By stop tightness: <1 ATR 39% (n=44); >=1 ATR 0% (n=2).
 - In dollars, real losses (-\$6,879) dwarf shakeouts (-\$1,447).
- **Expired option losses:** 20 of 60 were green at some point before the contract's own expiry. The 7-day plan window ends many 8–21 DTE contracts early.
- **Today's IWM 290P and 275P** shakeouts were Loss Rules **time stops**, not price stops. gex_scanner tags 16-DTE puts day.
- **SPY 770C and 768C** were correct stops.
- **LGHL** has not recovered to entry so far; its window is still open.

5.3 Stop-rule what-ifs

Net \$ on 264 closed trades:

Rule	All	H1	H2
Recorded	-4,070	-5,500	+1,429
Current, bar-priced	-5,097	-7,351	+2,253
Contract >=25%	-7,061	-8,586	+1,525
Contract >=35%	-4,217	-5,936	+1,719
1.25×ATR	-603	-3,359	+2,756

- **Contract-% stops: Not supported** (worse in both halves).
- **1.25×ATR: back to Measuring.** +\$5,108 of its gain is one MDB 350C trade published twice. Without it, the floor is -\$801 across 72 stop-outs.

- **The floor does not cover** `gex_scanner` **or** `gex_magnet`. 33 of 50 `gex_scanner` swing ideas since 09-24 have stops under 1.2 ATR.

5.4 Duplicates

- Any-strike duplicates (same symbol, side and engine, overlapping or same/next day): **59 ideas**. The 38 closed ones lost **-\$2,621** (-\$68.97 per trade), against -\$11.46 per trade for first instances. 21 are still open.
- Exact-instrument duplicates: 31 ideas, 15 closed, -\$1,218.
- Today's deploy adds same-instrument dedup only.

5.5 Exit-time stamping

- 129 distinct stamps for 522 exits. Three sweeps account for 221 rows:
 - Sep 24 9:15:51 ET: 103 rows;
 - Sep 8 2:43:12 PM: 95 rows;
 - Sep 1 12:41:18 PM: 23 rows.
- **The Sep 30 11:40 ET cluster** (CRM ×3, FNV, AMC, XBI):
 - The code path is `performance-validation-service.ts refineExitTiming -> shared/exit-hit-time.ts firstLevelTouch`, anchored at `server/lib/exit-time-bars.ts entryAnchorMs = max(publish, triggerObservedAt)`.
 - All six share `triggerObservedAt = 11:35:35 ET`, from one trigger-observer pass. The first bar after it (11:40) is "the hit" because price was already through the stop.
 - The real first crossings were CRM 9:35 ET, XBI **Sep 28 1:10 PM**, and FNV/AMC Sep 29 9:30.
- The validator's own stamp is `formatExitDate(now)` (`server/performance-validator.ts`, target and stop branches).
- Exit premiums remain the sweep quote.

5.6 Production memory restarts and in-memory state

- **pm2 restarts** (`max_memory_restart`):
 - Web: 700 MB limit, observed peaks up to 1.31 GB.
 - Worker: 900 MB limit, peaks up to 1.27 GB.
 - 40 restarts today, 20 of each. The worker restarted every 2–10 minutes between 17:15 and 20:11 UTC.
- **Effect on in-memory state:**
 - The outcome tracker starts 2 minutes after boot (`server/background-jobs.ts`), then runs every 5 minutes. With the worker living 2–10 minutes, most passes were first-after-boot sweeps. That is how exit stamps cluster.
 - The trigger observer (2-minute loop), dedup windows, GEX aggregates (60-second shared cache), journal mark caches and the heavy-job gate queue all reset on every restart.
 - Setups can be observed late: first pass after boot, as in 5.5.
 - Any intraday state that is not persisted (e.g. ORB ranges, intraday highs and lows between polls) is lost.
- **Research-run disclosure:**
 - One web restart (21:59:39 UTC) happened while this review's first read-only script was starting. The script is a separate process; its contribution to web memory cannot be excluded, and host memory was tight.
 - Later runs (to about 22:25 UTC) coincided with no restarts.

5.7 Deploy regression found during this review

- c1980ccb (deployed 17:10 CT) adds, in `server/journal-sources.ts` `loadDesk:`
`sql`substring(${tradeIdeas.outcomeNotes} from '\[exit-time:([a-z_]+)\]')``
- The template's cooked string drops the backslashes. Postgres receives `'[exit-time:([a-z_]+)']` and rejects it: `invalid regular expression: parentheses ( ) not balanced`.
- `loadDesk` throws, so **the NEXUS ideas book in the journal (and anything that calls `loadJournal(...'desk')`) fails** since the 22:12 UTC web restart.
- The fix is to escape the backslashes as `'\\[exit-time:\\\\]'` or pass the pattern as a bound parameter.
- The research scripts here load the desk directly through the same pure mapper to stay independent of this bug.

6. Replay evidence produced today

Supplied by the operator from `research/flush-reclaim-replay.ts`; not re-run here.

Hypothesis	Result	Status
Failed-squeeze short (flush then reclaim fails)	Held in both halves , about +0.2R per trade	Measuring. Positive out of sample, but n and costs not yet disclosed; paper-trade before promotion
Weekly flag breakouts	Only the version confirmed by volume drying up beat baseline on the +10%-in-4-weeks test. On real call prices it failed walk-forward	Not supported as an options strategy

These follow the signal-lab rule: a short-window win is a regime artifact until it survives a walk-forward on real contract prices.

7. Findings register

Owners: **Eng-Outcomes** = outcome tracker and trigger observer; **Eng-Pipeline** = idea producers and ingestion; **Model-GEX** = GEX engine and gex producers; **Bot** = Quantinum Bot; **Ops** = hosting and deploys; **Operator** = Malik.

ID	Finding	Severity	Owner	Remediation
F-0	Deploy c1980ccb breaks the desk journal loader (regex escaping in a sql template)	Critical	Ops / Eng-Outcomes	Escape the backslashes or bind the pattern; add a test that runs <code>loadDesk</code> against Postgres; redeploy outside 08:30–10:30 ET
F-1	Option stop exits priced at the tracker pass, not at the stop. 17 "hit_stop" rows show gains (+\$1,166)	High	Eng-Outcomes	Re-price at the stop bar (contract bar or model) when the hit time is known; flag "premium at sweep" rows and exclude them from stop statistics until repaired
F-2	Trigger observer triggers ideas whose stop is already breached. The exit-time repair then anchors the "hit" to the trigger pass (the Sep 30 11:40 cluster)	High	Eng-Outcomes	Refuse to trigger when price is beyond the stop; mark such ideas "invalidated before trigger" (no P&L); anchor hit time to the first real breach
F-3	new <code>Date(expiryDate)</code> parses date-only expiries at 00:00 UTC, so expiry-day options are "expired" at 8 PM ET the evening before (9 rows today, exits before publish)	High	Eng-Outcomes	Parse the expiry as 16:00 ET (or 16:15 for index products); re-grade the 9 rows; ODTE results are invalid until then
F-4	374 of 844 ideas are unscorable: <code>spx_session</code> and 43 <code>gex_scanner</code> options have no entry premium; 108 resolved without an exit premium	High	Eng-Pipeline	Block publish of an option idea without an entry premium; capture the exit premium at resolution or mark it unmeasured

ID	Finding	Severity	Owner	Remediation
F-5	83% of exit stamps are batch sweeps; hit time is unknown for most pre-09-30 exits	Medium	Eng-Outcomes	Run <code>scripts/repair-exit-dates.ts</code> (dry run first) with the F-2 anchor fix; keep the "hit time unknown" label (deployed today)
F-6	After-close option publishing: -\$4,516 (n=72) vs RTH +\$1,758 (n=73). 27 of 72 stops first crossed in the opening bar	High (\$)	Eng-Pipeline	Stage after-close ideas as candidates and publish or re-price after 10:00 ET; measure for 4 weeks
F-7	<code>gex_scanner</code> tags multi-week contracts <code>holding_period= 'day'</code> , so Loss Rules put a noon time stop on 16-DTE puts (the IWM 290P and 275P shakeouts)	Medium	Model-GEX	Derive the holding period from DTE; re-check time-stop coverage
F-8	<code>gex_scanner</code> and <code>gex_magnet</code> bypass the ingestion gates (ATR floor and others): 33 of 50 recent swing stops are under 1.2 ATR; <code>gex_scanner</code> is negative in both halves	Medium	Model-GEX / Eng-Pipeline	Route through <code>trade-idea-ingestion.ts</code> ; hold <code>gex_scanner</code> publication to "measuring" until a replay is positive
F-9	<code>market_scanner</code> call swings (8–21 DTE): -\$3,738 (n=86), concentrated in the Aug 31 batch	High (\$)	Eng-Pipeline	Suspend or re-gate pending a path replay on real call prices, with a halves split
F-10	Any-strike duplicates: 59 ideas, -\$2,621 closed; the copies lose about 6× as much per trade as first instances	Medium	Eng-Pipeline	Extend dedup to symbol + side + engine within a session, whatever the strike or expiry
F-11	The conviction band and probability band do not rank outcomes (S worst; B > A)	Medium	Eng-Pipeline (convictions)	Stop displaying the band as a quality signal until it is calibrated; refit on post-fix outcomes with a halves split
F-12	Bot Run 3 is 0/9 (-\$4,632); premium stops -\$9,080, although holding would have lost more	Medium	Bot	Restrict the bot to quant / band-A signals until the other sources replay positive; keep the stops
F-13	1.25×ATR floor not confirmed on this window (gain driven by one duplicated trade)	Low	Eng-Pipeline	Keep as "measuring"; re-test at n >= 150 stop-outs with duplicates removed
F-14	<code>options_flow_history.sentiment</code> is always 'unknown'; chain-snapshot flow has no direction	Medium	Eng-Pipeline (flow)	Derive the side from trade-vs-quote or drop directional flow claims; audit the "tape flipped" supersede logic that depends on it
F-15	Tradier key dead; premiums come from CBOE/Alpaca fallbacks without per-row source attribution	Medium	Operator / Ops	Renew the key; store <code>premium_source</code> with each entry and exit premium
F-16	40 memory restarts today; in-memory state (trigger observer, dedup, caches) reset every 2–10 minutes	High	Ops	Raise the limits to match the heap flags or cut the heap (GEX parse, chain loaders); persist trigger and dedup state; alert on more than 3 restarts per hour
F-17	Expired option losses: 20 of 60 were green before the contract's own expiry. The 7-day backstop shortens 8–21 DTE plans	Low	Eng-Outcomes	Align the backstop with the plan's stated horizon or DTE; report both

8. Limitations

- H2 has only 46 closed desk trades and 9 bot fills. Half-split conclusions are indicative only.
- Option re-pricing uses 1-hour trade-print closes from Alpaca's indicative feed. Illiquid contracts can be stale within the hour, and 3 option trades had no prints. No bid/ask or slippage is modelled.
- The desk book is unit-sized (1 contract or \$1,000). Dollar rankings over-weight high-premium contracts; per-trade % averages are in the results JSON.

- Pre-09-30 exit times are mostly sweep stamps. Hour-of-exit analyses are not reported for that reason.
- The ATR used is a 14-day simple average of true range from Yahoo daily bars, matching `trade-idea-ingestion.ts`.
- "Green" means the contract (or stock) closed above entry on at least one bar after the stop, within the plan window. It is not a claim that the trade was exitable at that price.
- The replay results in section 6 were not re-run here.

9. Ongoing monitoring

Re-run weekly and before any promotion:

- `research/loss-attribution.ts` and `research/shakeout-review.ts`: the halves split, duplicate counts and exit-stamp sharing (target: <10% shared stamps once F-2 and F-5 are fixed).
- The path replay (`research/path-replay.ts`) against the tracker: agreement rate.
- Coverage: scorable ideas as a share of published ideas (target >90% after F-4).
- pm2 restarts per day (target 0).

10. Post-report updates (2026-09-30 evening)

Changes and results after sections 1-9 were written. The deploys and replays below were done by the operator; figures are as reported and were not re-run for this report. Status labels follow the same rules as the body.

#	Update	Evidence	Status	Register
1	Journal regression fixed. The 18:11 ET deploy broke the NEXUS journal book for about 17 minutes (regex escaping in a sql template; section 5.7).	Fixed at 18:28 ET. Verified that the desk book loads all 470 rows.	Fixed	F-0 closed. The LoadDesk-against-Postgres test in its remediation remains the guard against a repeat.
2	Loss Rules v1.1 deployed 18:54 ET. Flow-led bot picks are exempt from the confluence rule and may enter until 15:00 ET.	The bot's own fills: single-source picks n=23, PF 1.67, +\$4,235, against picks the rule kept n=17, PF 0.51, -\$4,132.	Measuring	M3 / M9. Fitted on the same fills it is judged on, and n is small; it needs out-of-sample fills before it counts. Compare F-12.
3	Memory. CBOE loader, 60-second GEX cache, convictions base boards, GC after heavy jobs, jemalloc decay, staggered worker boot.	No memory restarts observed after the 18:11 ET deploy (worker 176-450 MB), against about 40 restarts earlier in the day.	Improving	F-16 stays open until a full regular session runs with 0 restarts.
4	ODTE classic-setup replay (last 12 months, SPY and single names, real option bars).	Only 'failed squeeze short, buy the nearest put, hold to close' survived both halves: +0.23 and +0.20 per \$1 (marginal). Lottos expire worthless 93-97% of the time.	Measuring	M4. Sniper enabled in watch mode in the 18:50s ET, publishing only that setup (at most 8 per day). Live grading still depends on F-3 (expiry-date parse).
5	Weekly bull-flag calls, replayed on real option prices.	Median +13.5% but average -3.8%; fails walk-forward.	Not supported	Confirms section 6: not an options strategy.
6	The evidence score does not rank outcomes.	S is the worst band per trade (-\$388.81, n=6); B beats A (+\$17.53 vs +\$13.58). Section 2 (M2) and Appendix A.1.	High	F-11 severity raised from Medium to High . Stop showing the band as a quality signal until it is recalibrated with a halves split.

Register after these updates: F-0 closed; F-11 High; F-16 improving but open; F-3 still blocks every ODTE result, including the sniper's. All other findings are unchanged.

Appendix A — Loss attribution, 2026-08-26 to 2026-09-30

The companion write-up (docs/LOSS_ATTRIBUTION_2026-09-30.md), reproduced in full. Section numbers below are the appendix's own; findings F-n refer to section 7 of the report.

Prepared 2026-09-30 (after the close). Read-only against production; nothing was written to the database.

- **Scripts:** `research/loss-attribution.ts` (P&L by dimension, duplicates, exit stamps) and `research/shakeout-review.ts` (post-exit paths, stop-rule what-ifs).
- **Results:** `research/loss-attribution-results.json` and `research/shakeout-review-results.json` (every number below is in them).
- **Window:** ideas published since 2026-08-26 (the clean-era baseline). Earlier outcomes are invalid because of known tracker bugs and are excluded.
- **Halves:** H1 = 08-26 -> 09-12 (218 closed trades), H2 = 09-13 -> 09-30 (46 closed trades). H2 is small, so treat its numbers as indicative only.

A. How the books are measured

- **NEXUS ideas (the desk book).** Every published idea is scored as a unit-sized trade, exactly as the journal scores it (`server/journal-row-maps.ts mapDeskIdea`):
 - options: 1 contract, at the recorded entry and exit premiums;
 - stocks and crypto: \$1,000 notional.
 - "\$ per trade" below means per unit-sized trade.
 - 470 ideas were scorable: 264 closed and 206 open. The open ideas are not in the P&L.
 - 374 ideas were excluded by the journal's own mapper rather than counted as \$0:
 - 158 expired with no measured exit;
 - 108 were options with no entry premium, which is **all 142** `spx_session` **ideas** plus 43 `gex_scanner` ideas;
 - 108 were resolved with no exit premium (102 of them `market_scanner`).
 - This coverage gap is itself a finding (see F-4 in the SR 11-7 v6 report).
- **Quantinum Bot.** Every `paper_positions` fill in every bot portfolio, at the bot's own size (`server/bot-ledger.ts`). That is 46 fills, of which 40 are closed.
- **Exit timing caveat.** The outcome tracker only ran continuously in production from 2026-09-30. Earlier exits were resolved in batch sweeps and carry the sweep time (see section 6). Two consequences:
 - Option exits recorded as `hit_stop` were priced at the contract quote **at the sweep**, not at the stop. This is why 17 `hit_stop` rows show a *positive* P&L (+\$1,166 in total).
 - `research/shakeout-review.ts` re-prices every stop from bars as an independent check.

A. 1. Where NEXUS loses money

Whole book: 264 closed trades. 103 won and 151 lost, a **39.0% win rate**. Net **-\$4,070**, or -\$15.42 per trade. The average win was +\$238.58 and the average loss -\$189.70.

Biggest \$ drains, ranked

These groups overlap, so their totals are not additive.

#	Drain	n	Win rate	Net \$	\$/trade
1	Ideas that hit their stop (all engines)	72	23.6%	-7,160	-99.44
2	Aug 31 batch : ideas published Mon Aug 31, mostly after the close, then marked in the Sep 1 and Sep 8 sweeps	74	32.4%	-7,634	-103.16
3	Options published after the close (16:00–24:00 ET)	72	34.7%	-4,516	-62.72
4	market_scanner · call · swing (mostly 8–21 DTE)	86	29.1%	-3,738	-43.47
5	gex_scanner (all) — negative in both halves (H1 -1,134, n=5; H2 -693, n=23)	28	25.0%	-1,827	-65.25
6	Duplicate ideas: same symbol, side and engine, any strike (section 5)	38 closed	28.9%	-2,621	-68.97
7	quant · stock · swing	8	12.5%	-1,013	-126.58

Losses are concentrated:

- The 15 worst trades account for -\$16,454, which is 57% of all gross losses (-\$28,644).
- 13 of those 15 are market_scanner calls published Aug 31 – Sep 1, expiring Sep 11 or Sep 18.
- 9 of the 15 ended "expired" at 7 days, between -15% and -99% (MDB 500C -\$2,120, NSC 310C -\$1,350, CBOE 250C -\$1,155, ...).

By contrast, options published during regular hours (RTH) made money: n=73, 37.0% win rate, **+\$1,757.50**.

By engine

Engine	n	Win rate	Avg win	Avg loss	Net \$	\$/trade
market_scanner	162	37.7%	294.31	-225.71	-3,941	-24.33
gex_scanner	28	25.0%	72.64	-111.22	-1,827	-65.25
zero_dte_desk	3	33.3%	20.00	-46.00	-72	-24.00
quant	55	47.3%	135.85	-134.95	+293	+5.33
flow	8	37.5%	557.00	-249.00	+675	+84.38
crypto_engine	5	40.0%	73.44	-29.55	+58	+11.65
gex_magnet / hybrid / tradingview	1 each	—	—	—	+13 / +210 / +520	—

spx_session does not appear: all 142 of its ideas were excluded for having no entry premium. sector_ignition (2 ideas) and premarket_gap (0 ideas) have no closed trades.

By asset, option side, holding period and DTE

Stocks and crypto are sized at \$1,000, options at 1 contract.

Slice	n	Win rate	Avg win	Avg loss	Net \$	\$/trade
Option	161	37.3%	374.93	-275.37	-2,563	-15.92
Stock	96	41.7%	43.75	-60.89	-1,660	-17.29
Crypto	7	42.9%	109.57	-44.11	+152	+21.75
Calls	136	36.0%	449.33	-309.97	-1,850	-13.61
Puts	25	44.0%	43.50	-85.07	-712	-28.50
Swing	208	39.4%	257.91	-221.66	-5,007	-24.07
Day	42	33.3%	96.80	-39.12	+299	+7.12
Position	14	50.0%	295.78	-238.74	+638	+45.57

Slice	n	Win rate	Avg win	Avg loss	Net \$	\$/trade
ODTE	15	40.0%	144.00	-18.67	+696	+46.40
8–21 DTE	112	34.8%	488.05	-323.48	-2,316	-20.67
22–60 DTE	28	46.4%	198.31	-275.92	-733	-26.18
Lotto (trade type)	20	55.0%	223.64	-157.56	+1,042	+52.10
Scalp (trade type)	20	25.0%	66.40	-38.14	-202	-10.10

By direction, day of week, entry time and conviction band

Slice	n	Win rate	Net \$	\$/trade
Long thesis	238	38.7%	-3,349	-14.07
Short thesis	26	42.3%	-722	-27.75
Monday publishes (all Aug 31)	74	32.4%	-7,634	-103.16
Tuesday	82	39.0%	+958	+11.69
Wednesday	86	45.3%	+2,690	+31.28
Published after close	123	38.2%	-4,973	-40.43
Published during RTH	122	38.5%	+747	+6.12
Published pre-market	19	47.4%	+156	+8.20
Conviction band S	6	33.3%	-2,333	-388.81
Conviction band A	41	39.0%	+557	+13.58
Conviction band B	85	31.8%	+1,490	+17.53
Conviction band C	103	38.8%	-5,280	-51.26
No band	29	62.1%	+1,495	+51.57

The conviction band does not rank outcomes. S, the top band, is the worst per trade, and B beats A on dollars. probabilityBand shows the same disorder: A -\$7,002 (n=130) against A- +\$7,026 (n=17).

By stop distance (underlying)

Stop distance	n	Win rate	Net \$	\$/trade
<0.5%	20	40.0%	+644	+32.20
0.5–1%	14	21.4%	-864	-61.74
1–2%	78	29.5%	-7,387	-94.71
2–4%	72	44.4%	+8,239	+114.43
4%+	80	46.3%	-4,702	-58.77

A. 2. Where the Quantinum Bot loses money

Whole book: 40 closed fills (all options; 39 calls, 1 put). 12 won and 28 lost, a 30.0% win rate. Net **+\$103**, or +\$2.58 per fill. The average win was +\$1,241.83 and the average loss -\$528.54.

Slice	n	Win rate	Net \$	\$/fill
Run 2 (Aug 26 – Sep 18)	30	40.0%	+4,908	+163.60
Run 3 (Sep 24 –)	9	0%	-4,632	-514.67
Exit: hard or soft stop	13	0%	-9,080	-698.46
Exit: trailing stop	4	0%	-1,020	-255.00

Slice	n	Win rate	Net \$	\$/fill
Exit: time / expiry	4	50%	+4,306	+1,076.50
Exit: other (gap_magnet, flow_reversal)	19	52.6%	+5,897	+310.37
Engine gex_scanner	3	0%	-1,967	-655.67
Engine market_scanner	15	26.7%	-1,127	-75.13
Engine flow	2	0%	-1,026	-513.00
Engine quant	20	40.0%	+4,223	+211.15
DTE 22–60	7	28.6%	-1,738	-248.29
DTE 60+ (SNOW 420C Jan-27)	1	0%	-1,538	-1,538.00
DTE 8–21	30	33.3%	+4,045	+134.83
Band S / B / C	9 / 10 / 11	33 / 10 / 18%	-2,220 / -1,689 / -1,196	—
Band A	10	60%	+5,208	+520.80
Friday entries	14	7.1%	-5,050	-360.71

The bot's drains, ranked:

1. Premium stops at -31% to -95%: -\$9,080.
2. Every Run 3 fill (from Sep 24) lost: -\$4,632. These were mostly gex_scanner, flow and market_scanner signals taken Sep 24–25.
3. Long-dated contracts (22+ DTE): -\$3,276.

Only quant-sourced signals and band A made money.

A. 3. Shakeouts: stopped out, then green

Method (research/shakeout-review.ts):

- For each closed losing trade, the script finds the first 5-minute regular-session bar that crossed the stop. Stocks use Yahoo bars; options use Alpaca 1-hour contract bars. For ideas that expired rather than hit their stop, the exit time is used instead.
- It then looks at the rest of the original plan window. That window is exit_by, else publish + 7 days, capped at the contract's expiry and at now.
- Each loss is labelled:
 - **shakeout -> target**: the underlying reached the original target after the stop;
 - **shakeout -> green**: the contract (or the stock) closed above entry at some point, but the target was not reached;
 - **real loss**: neither happened.

Population	n	Real loss	-> target	-> green	Shakeout rate	Lost \$
All closed losses	151	121	10	20	19.9%	-28,645
Stop-outs (hit_stop)	46	29	6	11	37.0%	-8,326
Expired or time-stopped losses	105	92	4	9	12.4%	-20,318

The dollars are in the real losses, not the shakeouts.

- The 17 stop-out shakeouts cost -\$1,447 in total. The 29 real stop-out losses cost -\$6,879.

- On the 36 stop-outs that could be priced, holding to the end of the plan would have lost **-\$5,081**, against -\$4,411 realised. The stops saved \$670 net.
- Of 60 expired option losses, 20 closed above entry at some point before the contract's own expiry. The 7-day plan window cut them off. This is a horizon mismatch, not a stop problem (see fixes).

Stop-out shakeouts, broken down:

Slice	n	Shakeout rate	Lost \$
quant	12	66.7%	-824
flow	3	66.7%	-992
gex_scanner	12	25.0%	-2,230
market_scanner	16	25.0%	-4,191
crypto_engine	3	0%	-89
Stop <1% of underlying	10	40.0%	-731
Stop 1–2%	20	45.0%	-2,564
Stop 2–4%	12	33.3%	-3,394
Stop 4%+	4	0%	-1,637
Stop <0.5 ATR	19	36.8%	-3,952
Stop 0.5–1 ATR	25	40.0%	-2,938
Stop >=1 ATR	2	0%	-1,436
Option down <10% at the stop bar	8	37.5%	-1,039
Option down 10–25%	7	14.3%	-1,881
Option down 35–50%	4	25.0%	-2,565
Option down >=50%	5	20.0%	-1,794
H1	30	46.7%	-6,069
H2	16	18.8%	-2,257

Today's examples, checked against bars:

- **IWM 290P and 275P (Oct 16, gex_scanner).** These were *not* price stops.
 - They hit the Loss Rules v1 **time stop**. gex_scanner tagged these 16-DTE puts holding_period='day', so horizonTradingDays gave a 1-day horizon.
 - Half of one day means an exit at 12:45 ET unless the trade was at least +0.5R.
 - They closed at -0.7% (-\$8) and -4.7% (-\$13.50). Both contracts later closed above entry (290P high 12.60 vs 11.51 entry; 275P 3.23 vs 2.865). Confirmed shakeouts.
- **IWM 280P (Oct 1).** Expired at its same-day deadline at -7.3% (-\$12). The contract later printed 2.20 vs 1.65 entry. Confirmed shakeout.
- **IWM 279P x3, IWM 280P (Sep 30), SPY 767C/768C (Sep 30), TSLA 355C/352.5C, META 732.5P.**
 - Their exits are stamped **Sep 29 8:00 PM ET, before they were published.**
 - Cause: server/performance-validator.ts (option-expiry check, ~line 702) does new Date(idea.expiryDate). A date-only string parses as 00:00 UTC, which is 8 PM ET the previous evening.
 - So any idea on an option that expires that day is "expired" at the first tracker pass and priced at whatever that pass reads. There are 9 such rows; they are 0DTE ideas and expiry-day holds.

- Two of the IWM 279P rows later reached the original target.
- **LGHL (quant, stock).** Stopped at 1:35 PM at -14.9% (-\$149). The stop was 14.9% away (0.54 ATR). **Not green so far:** the best 5-minute close after the stop was still 2.9% below entry. Its 7-day window is still open.
- **SPY 770C and 768C (Oct 1).** Correctly stopped. Neither recovered; horizon values were 0.29 vs 1.60 and 0.58 vs 1.78.

A. 4. Stop-rule what-ifs (same closed trades)

The rules compared:

- **A** — the current rule, re-priced at the first bar that crossed the underlying stop.
- **B** — exit only when the underlying stop is through **and** the contract is down at least 25%.
- **C** — the same as B at 35%.
- **D** — the stop is never tighter than $1.25 \times \text{ATR}(14)$, for swing and position ideas.

Only the 72 `hit_stop` rows can change. The other 192 closed trades are identical under every rule, so they are carried at their recorded P&L. Where a variant cannot be priced from bars (11–15 rows), the recorded P&L is kept.

Rule	All 264: net \$	Win rate	H1 (218) net \$	H2 (46) net \$
Recorded	-4,070	39.0%	-5,500	+1,429
A current, bar-priced	-5,097	39.4%	-7,351	+2,253
B option $\geq 25\%$	-7,061	36.7%	-8,586	+1,525
C option $\geq 35\%$	-4,217	37.9%	-5,936	+1,719
D $1.25 \times \text{ATR}$ floor	-603	39.8%	-3,359	+2,756

Robustness check: the change against A on the 72 stop rows.

Rule	Change in \$ vs A	Trades better / worse	Change in \$ excluding the two duplicate MDB 350C rows
B	-1,962	6 / 25	-7,070
C	+718	9 / 26	-4,390
D	+4,307	21 / 27	-801
D, H2 only	+503	5 / 7	(driven by DELL +\$890)

What the what-ifs show:

- **Contract-% stops (B, C) make things worse in both halves.** They keep losers open until the contract is already down 25–35%.
- **The ATR floor (D) looks best, but its whole gain comes from one trade published twice.** MDB 350C, published Aug 25 9:05 PM and again Aug 26 4:35 AM, gapped up after the stop; that is +\$5,108.
 - Without those two rows, D is flat to slightly negative: -\$801 over 72 stops, with 21 trades better and 27 worse.
 - The v3 walk-forward that supported the floor ($n=332$) is **not confirmed on this window**. The label stays "measuring".
- **Does the $1.25 \times \text{ATR}$ floor already apply to swings? Only for engines that write through `server/trade-idea-ingestion.ts`.**
 - `gex_scanner` (`storage.createTradeIdea` in `server/gex-idea-scanner.ts:366`) and `gex_magnet` (`server/gex-magnet-actions.ts:138`) bypass it.

- Swing and position ideas published since 2026-09-24 with a stop tighter than 1.2 ATR:

Engine	Below 1.2 ATR
gex_scanner	33 of 50
gex_magnet	1 of 1
quant	2 of 2
market_scanner	25 of 141

- **Stop timing:**

- 27 of 72 stops were first crossed in the **9:30 opening bar**: ideas published after the close that gapped through the stop.
- 9 of 72 stops are **not crossed in any regular-session bar**. Examples are NVDA recorded at 4:45 PM and GOOGL at 4:22 PM. They were triggered on extended-hours or stale quotes.

A. 5. Duplicates

Definition	Groups	Duplicate ideas	Closed	Win rate	Net \$	\$/trade	Still open
Same symbol + side + engine + same instrument , overlapping or same/next ET day	21	31 (26 while an earlier one was open)	15	33.3%	-1,218	-81.19	16
Same symbol + side + engine, any strike or expiry	28	59	38	28.9%	-2,621	-68.97	21

- For comparison, first instances returned -\$11.46 per trade. The second and later copies lose about 6–7× as much per trade.
- By engine (any-strike, closed):

Engine	n	Net \$	Wins
gex_scanner	15	-665	2
quant	16	-598	—
market_scanner	3	-1,377	—

- Examples:
 - IWM 279P ×3 (Sep 30);
 - XBI 150P ×5 (Sep 26–29);
 - QQQ 700P and AVGO 400C ×4 each;
 - CRM 220P ×2 (both -\$232);
 - WDC 530C ×3 within 40 minutes on Aug 26 (-\$824 and -\$220).
- 87 further rows were already collapsed in the database (resolution_reason='duplicate_collapsed').
- A same-instrument dedup (server/lib/instrument-dedup.ts) was deployed at 17:10 CT today. It does not cover the any-strike case.

A. 6. Exit-time sanity

- Of the 522 closed ideas that have an exit_date, there are only **129 distinct timestamps**.

- **49 timestamps are shared by two or more different symbols. They cover 433 rows (83%).**
- Only 33 rows carry an [exit-time:...] tag: 14 bar_hit, 17 deadline, 2 live. 76 rows have a date-only stamp.

Stamp (ET)	Rows	Symbols	What
Sep 24 9:15:51 AM	103	103	one sweep; all auto_expired (market_scanner, quant)
Sep 8 2:43:12 PM	95	95	one sweep; all auto_expired
Sep 1 12:41:18 PM	23	23	one sweep; 18 stops, 3 targets, 2 expired
Sep 29 8:00:00 PM	12	5	the UTC-midnight expiry bug above; deadline tag
Sep 30 11:40:00 AM	6	4	CRM x3, FNV, AMC, XBI — bar_hit, "repaired from sweep stamp 10:42 CT"

- **Code path that sets the exit time.**
 - server/performance-validator.ts stamps target and stop hits with exitDate: formatExitDate(now): the tracker pass time, in America/Chicago.
 - server/performance-validation-service.ts refineExitTiming() (added today in c896300b) then replaces that with the first bar at or after the **entry anchor** that crossed the level (shared/exit-hit-time.ts firstLevelTouch).
 - The entry anchor is server/lib/exit-time-bars.ts entryAnchorMs: the later of the publish time and executionAudit.triggerObservedAt.
 - The exit premium is still the quote at the pass. The note says so: "contract exit premium unchanged (quoted at the sweep)".
- **The Sep 30 11:40 AM cluster is one trigger-observer pass, not six stops in one bar.**
 - All six rows have triggerObservedAt = 2026-09-30T15:35:35.404Z.
 - The first 5-minute bar after that is 11:40. Each underlying was already beyond its stop at 11:40, so each one "crossed" in that bar.
 - Independent bars show the real first crossings:
 - CRM (stops 227.43 and 228.14): the 9:35 AM bar;
 - XBI (157.36): **Sep 28 1:10 PM**;
 - FNV and AMC: the Sep 29 9:30 opening bar.
 - Two errors follow:
 1. The trigger observer marked ideas "triggered" whose stop was already violated.
 2. The repair anchored the hit time to that trigger rather than to the real breach.

A. Recommended fixes, ranked by \$ impact in this window

1. **Stop publishing options after the close, or hold them until the next session's first 30 minutes and re-price.**
 - After-close options lost -\$4,516 (n=72) while RTH options made +\$1,758 (n=73).
 - 27 of 72 stops were first crossed in the opening bar.
 - Owner: idea pipeline (market_scanner scheduler).
2. **Suspend or re-gate market_scanner call swings (8–21 DTE) until a walk-forward replay shows a positive half.**
The group lost -\$3,738 on n=86, and the Aug 31 batch alone lost -\$7,634.

3. **Fix the stop-out bookkeeping before trusting any stop metric.**

- Price option stop exits at the stop bar, not at the sweep: 17 "stops" are booked as wins (+\$1,166).
- Do not trigger an idea whose stop is already breached.
- Anchor the repaired hit time to the first real breach, not to `triggerObservedAt`.

4. **Fix the expiry-date parse** in `server/performance-validator.ts`. Treat `expiryDate` as 16:00 ET on that date. Otherwise every 0DTE or expiry-day idea is "expired" before it is published (9 rows today; all 0DTE results are invalid until fixed).

5. **Collapse any-strike duplicates** (same symbol, side and engine within a session). That is 59 ideas, -\$2,621 closed, 21 still open.

6. **gex_scanner and gex_magnet:**

- route them through the ingestion gates (the ATR floor);
- stop tagging multi-week contracts `holding_period='day'`, which puts a noon time stop on 16-DTE puts (the IWM 290P and 275P shakeouts);
- `gex_scanner` lost in both halves (-\$1,827, 25% win rate).

7. **Do not adopt contract-% stops (B or C).** They were worse in both halves.

- Keep the ATR floor as "measuring": this window neither confirms nor refutes it once the duplicated MDB trade is removed.
- Revisit with $n \geq 150$ stop-outs.

8. **Bot:**

- Premium stops cost -\$9,080. Holding those contracts to the end of the plan would have cost more (-\$14,268 vs -\$11,340 realised on the 25 that could be priced), so the stops are not the leak. Signal selection is.
- Restrict the bot to quant-sourced / band-A signals (+\$4,223 and +\$5,208) until the Run 3 engines (`gex_scanner`, `flow`, `market_scanner`: 0/9) show a positive replay.